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$$(-3, -11, -17, -15, 1, 37, 99, \dots)$$

$$\Rightarrow (-8, -6, 2, 16, 36, 62, \dots)$$

$$\Rightarrow (2, 8, 14, 20, 26, \dots)$$

$$\Rightarrow (6, 6, 6, 6, \dots)$$

$$x_n = an^3 + bn^2 + cn + d$$

$$\Rightarrow \begin{cases} a + b + c + d & = & -3 \\ 8a + 4b + 2c + d & = & -11 \\ 27a + 9b + 3c + d & = & -17 \\ 64a + 16b + 4c + d & = & -15 \end{cases} \Rightarrow \begin{cases} 7a + 3b + c & = & -8 \\ 19a + 5b + c & = & -6 \\ 37a + 7b + c & = & 2 \end{cases} \Rightarrow \begin{cases} 12a + 2b & = & 2 \\ 18a + 2b & = & 8 \end{cases}$$

$$\Rightarrow 6a = 6 \Rightarrow a = 1$$

$$\Rightarrow b = -5 \Rightarrow c = 0 \Rightarrow d = 1$$

$$x_n = n^3 - 5n^2 + 1$$

$$x_8 = 8^3 - 5(8^2) + 1 = 512 - 320 + 1 = 193$$

References